

Yumeng He

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EDUCATION

University of Southern California

Aug 2024 – May 2026

Master of Science

Los Angeles, CA, USA

- Cumulative GPA: 3.83/4.0
- Computer Science major
- Selected Coursework: Computer Graphics, Computer Animation and Simulation, 3D Graphics and Rendering

University of Toronto

Sep 2019 – May 2024

Honours Bachelor of Science with High Distinction

Toronto, ON, Canada

- Cumulative GPA: 3.71/4.0
- Computer Science and Mathematics double major
- Selected Coursework: Calculus I & II, Linear Algebra I & II, Ordinary Differential Equations, Abstract Mathematics, Complex Variables, Number Theory, Machine Learning, Artificial Intelligence

PUBLICATIONS

- Zhenlian Miao, Guangzhu Chen, Xiaojuan Liao, Jiu Dai, **Yumeng He**. *Self-supervised dual-layer 2D normalizing flow method for industrial anomaly detection*. Applied Soft Computing (ASOC), 2024.

RESEARCH EXPERIENCE

USC RESL Lab Research Assistant | Part-time

July 2025 – present

Supervisor: Prof. Gaurav S. Sukhatme

Los Angeles, CA, USA

- Ledding the Real2Sim pipeline for a robotic manipulation project, focusing on building a digital twin from a single RGB-D image of a cluttered tabletop scene—including object segmentation, mesh generation, and estimation of pose, scale, and placement—and importing the reconstructed scene into ManiSkill3 for simulation

UCLA AIVC Lab Visiting Student | Full-time

June 2025 – present

Supervisor: Prof. Chenfanfu Jiang

Los Angeles, CA, USA

- Conducting full-time research on 3D articulated object generation, with a focus on neural representations of shape and joint dynamics for physically plausible synthesis
- Exploring physics-aware scene layout and hierarchical composition, combining structural priors with simulation constraints to guide object placement
- Preparing a first-author submission to CVPR 2026, presenting a unified framework for articulated object synthesis guided by both data-driven learning and physical consistency

CDUT Research Assistant | Full-time

May 2023 - Aug 2023

Supervisor: Prof. Guangzhu Chen

Remote

- Faced with unlabeled industrial defects on MVTec AD and tasked to raise unsupervised detection/localization accuracy, proposed SS-DualFlow: a dual-layer 2D normalizing-flow that maps features to a Gaussian base to curb information loss and inserts an Exponential Space Attention module to focus on anomaly-salient regions. Result: image-level AUROC = 99.38% and pixel-level AUROC = 98.38% on MVTec AD; co-authored the Applied Soft Computing (2024) paper.

RELEVANT PROJECTS

Incompressible Fluid Simulation: A Comparison | C++, OpenGL

Spring 2025

- Implemented and benchmarked four 2D incompressible fluid solvers, including Stable Fluids, Smoothed Particle Hydrodynamics (SPH), Particle In Cell (PIC) and Affine Particle In Cell (APIC) on identical scenarios, reporting performance-vs-accuracy trade-offs.

Collision Detection with Penalty Method and IPC | C++

Spring 2025

- Built a physically accurate 3D jello cube simulator using a mass-spring system with structural, shear, and bend springs to model real-world deformation under force
- Implemented penalty-based collision detection to handle interactions with static obstacles—including inclined planes and spheres—ensuring realistic response under contact
- Upgraded basic collision logic with Incremental Potential Contact (IPC), reducing interpenetration to below $1e-4$ relative gap across 1,000+ simulation steps, effectively achieving zero visible penetration and stable behavior

Inverse Kinematics with Skinning | C++

Spring 2025

- Implemented Linear Blend Skinning and Dual Quaternion Skinning to deform 3D characters with smooth and realistic joint articulation
- Built a full FK–IK pipeline, supporting both forward kinematics for pose propagation and inverse kinematics for pose estimation under joint constraints
- Developed multiple IK solvers, including the pseudoinverse method (least squares) and Tikhonov regularization (damped least squares) to improve stability

Motion Capture Interpolation | C++

Spring 2025

- Implemented four interpolation schemes (incl. linear & Bézier) with both Euler- and quaternion-based rotations to reconstruct and smooth optical mocap, mitigating gimbal lock and interpolation artifacts
- Generated natural, continuous transitions across keyframes for high-dimensional skeletal animation

Ray Tracing | C++, Rendering

Fall 2024

- Built a modular ray tracer in C++ supporting both triangle meshes and spheres, featuring Phong shading, shadow rays, and recursive reflections for realistic mirror effects
- Integrated supersampling antialiasing, soft shadows via area lights, and the Moller–Trumbore algorithm for fast ray–triangle intersections
- Achieved visually convincing renders with enhanced realism and performance, balancing image quality and computational cost across complex scenes

Simulating a Roller Coaster | OpenGL, Rendering

Fall 2024

- Built a first-person roller-coaster sim with Catmull–Rom spline tracks and a Frenet-frame on-rail camera; added speed control, auto-screenshots, and seamless segment transitions.

INDUSTRY EXPERIENCE

Software Developer Intern | Full-time

Aug 2022 – Aug 2023

HCL Canada Inc.

Toronto, ON, Canada

- Automated build and deployment pipelines using Jenkins, accelerating production releases and minimizing manual errors by eliminating key bottlenecks in CI/CD
- Containerized microservices for both x86_64 (xLinux) and IBM PowerPC (pLinux) architectures, enabling architecture-agnostic builds and scalable, consistent deployments
- Integrated SonarQube into the pipeline; documented a detailed usage guide, presented to cross-functional teams, and resolved 3 critical bugs and 308 code smells (including 4 blockers, 41 major issues)
- Deployed a shell script to auto-clean legacy logs weekly (Sunday 00:00 EST), reducing disk clutter and improving system performance with zero manual intervention

COMMUNITY SERVICES

- Mentor for Viterbi Graduate Mentorship Program
- Mentor for Women in Engineering Mentorship
- Mentor for Viterbi Graduate Mentorship Program

Fall 2025

Fall 2025

Summer 2025

INVITED TALKS

- MPC Is All You Need.* [Co-presented]. USC SLURM Lab, hosted by Prof. Daniel Seita

Aug 2025

AWARDS & HONORS

- Dean’s List Scholar
- Ranked in the top 25% of contestants in the Galois Contest
- Ranked in the top 25% of contestants in the Cayley Contest
- Ranked in the top 25% of contestants in the Canadian Intermediate Mathematics

2020 - 2024

2017

2017

2016

TECHNICAL SKILLS

- Operating Systems:** Linux, Windows, MacOS
- Languages:** Mandarine, English, C++, Python, HTML, CSS, JavaScript, MySQL, R
- Graphics and Simulation:** OpenGL, GLSL, Eigen, PyTorch
- Tools and DevOps:** Docker, Jenkins, Gradle, Ant, Shell, Git, Github Actions